

Uzair Abid

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Research Interests

I am passionate about **Natural Language Processing**, **Agentic AI**, and **Computer Vision**. My interests revolve around developing intelligent systems that can understand language, learn from data, and interact with the world in meaningful ways. I enjoy combining theory with practical applications and continue to explore research that bridges academic ideas with real world use cases.

Currently, along with my work, I'm engaged in an ongoing research project with Dr. Boram Lim (former Prof. @ University of Kansas) on using Large Language Models to study consumer behavior, specifically exploring how different review algorithms affect restaurant selection decisions through AI agent simulations.

Professional Experience

Head of Web Data Acquisition, Wealthica – Canada (Remote) Jun 2025 – Present

Software Engineer Jul 2024 – Jun 2025

- Engineered robust backend systems using **Node.js**, **TypeScript**, **MongoDB**, **Redis**, **Docker**, and **REST APIs**, powering scalable financial data pipelines at Wealthica.
- Designed and implemented an AI-powered auto-remediation platform using **Claude API**, **GitHub Actions**, **RabbitMQ**, **SigNoz**, **Slack**, and **ClickUp** to detect connector failures, scrub PII, generate draft PRs, and apply path-restricted AI fixes through CI/CD workflows, reducing manual triage time by **70%** and accelerating issue resolution by **50%**.
- Designed and developed an AI-powered browser automation platform using a custom-trained ML model to analyze web pages and automate data extraction, improving extraction efficiency by **35%** and reducing manual effort by **60%**.
- Mitigated **30%** request failures during peak hours by analyzing and implementing pre auto-scheduling with **AWS CDK**.
- Helped fintech clients like **Questrade**, **Flinks**, and **white-label partners** in API integration of Wealthica with secure auth flows and optimized data access.

Software Engineer (ML/AI & Backend), AutomateBoring – United States (Remote) Jul 2023 – Jul 2024

- Built **retrieval-augmented generation (RAG)** systems pairing **embedding models** with **vector databases** (Pinecone, FAISS) to ground LLM outputs in proprietary data, substantially reducing hallucinations on knowledge-intensive tasks.
- Fine-tuned open-source **LLMs** using **LoRA/QLoRA** (PEFT) on curated instruction datasets, reaching quality comparable to far larger models while keeping training and inference costs low.
- Accelerated model inference through **quantization** (4-bit/8-bit) and **vLLM**-based serving, increasing throughput while cutting GPU memory footprint and latency in production.
- Developed **computer vision** pipelines for object detection, segmentation (**YOLOv8**), and **OCR** in **PyTorch**, using mixed-precision and distributed (**DDP**) training to speed up experimentation.
- Established **MLOps** practices including experiment tracking (**Weights & Biases**), automated evaluation harnesses (with **LLM-as-judge**), model versioning, and **CI/CD** deployment, enabling reproducible experiments and dependable releases.

AWS DevOps Engineer, MyAskAI – United Kingdom (Contract - Remote) Apr 2023 – Jul 2023

- Analyzed application performance using **Datadog** and **Cloudwatch**, detecting **15 issues** that were promptly addressed, leading to a **25%** reduction in user-reported incidents.
- Successfully migrated a three-tier app from Render to AWS with CI/CD using **AWS CodePipeline**, improving performance by **30%** and reducing hosting costs by **20%**.
- Deployed a machine learning model on AWS, building an end-to-end pipeline with model hosting, input preprocessing, and result handling using services like **Lambda**, **S3**, and **API Gateway**.

AWS DevOps Engineer, Rehani Soko – United States (Contract - Remote) Jan 2023 – May 2023

- Developed and optimized microservices for a real estate application, integrating AWS services such as serverless **Lambda**, **S3**, and **DynamoDB** to improve scalability and reduce latency by **18%**.
- Implemented CI/CD pipelines using AWS **CodePipeline** and **CodeBuild**, enabling automated testing, deployment, and rollback across multiple environments, cutting release times by **40%**.

Education

Sir Syed University of Engineering and Technology, BS in Computer Engineering Oct 2020 – Jul 2024

- CGPA: 3.58/4.0
- Relevant Coursework: Artificial Intelligence, Operating Systems, Database Management Systems (DBMS), Software Engineering, Object Oriented Programming (OOP)

Publications & Preprints

Using Large Language Models to Study Consumer Behavior: Review Algorithm Effects on Restaurant Selection Co-Author · In-Progress

Smart Shelf Advertising using Real-Time Product Segmentation and Interaction on Low-Cost Edge Devices Co-Author · Preprint

Research Experience

Using Large Language Models to Study Consumer Behavior: Review Algorithm Effects on Restaurant Selection Co-Author · In-Progress

Working with **Professor Boram Lim** (former University of Kansas) on a research project that uses AI agents powered by GPT models to simulate realistic customer decision-making in restaurant choice scenarios. The study explores how different review presentation algorithms (highest rated vs. most recent) and restaurant positioning (high-end vs. casual) affect consumer behavior. The system includes intelligent bias detection, dynamic review supplementation, and detailed behavioral logging.

Smart Shelf Advertising using Real-Time Product Segmentation and Interaction on Low-Cost Edge Devices Co-Author

Developed a real-time retail marketing system that intelligently detects product interaction and displays contextual advertisements on low-cost hardware. The system integrates Apriltag tracking, YOLOv8-based object segmentation, and Arduino-controlled lighting to detect when a product is picked and trigger the corresponding video ad. A custom YOLOv8 segmentation model trained on 3500+ images achieved **97.4% mAP@0.5**, demonstrating robust real-time performance in retail deployments.

Simulation Study on Edge Cloud Traffic Research Assistant

Assisted **Engr. Muhammad Rehan Rasheed** in developing a simulation project in **Java**, focusing on traffic management and performance analysis for edge cloud environments.

Teaching Experience

Stanford's Code in Place

Section Leader

Taught basic Python to 10+ students from all over the world and guided them through Stanford's CS106A course.

Completion Certificate: [View Certificate](#)

ICodeGuru

Trainer & Moderator

At ICodeGuru, we teach, train, mentor, and guide students from underprivileged areas of Pakistan to improve their professional skills. I taught coding to these students and helped them solve LeetCode problems.

Volunteer

North Dakota State University

Assisted Dr. M. Zubair Malik in modernizing **CSCI 717 – Software Construction** taught at NDSU, integrating AI driven development techniques and advanced cloud computing concepts, including AWS, to align the course with industry standards.

Projects

Resume AI Assistant

[\[Code\]](#) · [\[View Demo\]](#)

Developed an intelligent resume optimization system powered by OpenAI GPT-4 and Pydantic for structured data validation. Features AI-powered resume tailoring against job descriptions, comprehensive gap analysis, automatic cover letter generation with customizable tones, visual diff highlighting of changes, and PDF upload support for seamless document processing.

PDF Q&A Bot with Multi-LLMs

[\[Code\]](#) · [\[View Demo\]](#)

Developed an intelligent document analysis system using multiple open-source language models (Phi-3.5, Qwen 2.5, Llama 3.2, Gemma 2) from Hugging Face. Features 4-bit quantization for memory efficiency, automatic CPU offload, and a user-friendly interface for uploading PDFs and asking context-aware questions.

AI Tutor - Adaptive Learning Assistant

[\[Code\]](#) · [\[View Demo\]](#)

Built an intelligent tutoring system using OpenAI GPT-4o-mini that adapts explanations to different complexity levels (1-6). Features real-time streaming responses and dynamic level adjustment for the explanations.

AI Calorie Tracker

[\[Code\]](#) · [\[View Demo\]](#)

Developed an intelligent food analysis application using OpenAI Vision API and Gradio. The system automatically identifies food items from the uploaded images and provides detailed nutritional information, including calories, fat, and protein content.

BlogPilot AI - WordPress Automation using CrewAI Agents

[\[Code\]](#) · [\[View Demo\]](#)

Developed an intelligent multi-agent content creation system using CrewAI that automates the entire blog writing workflow. Features four specialized AI agents for research, outlining, writing, and publishing, with seamless WordPress integration via REST API for automated content publishing.

Analysis of Media Posts

[\[Code\]](#)

Designed an NLP pipeline using scikit-learn to classify and analyze mental health related social media posts. The model identifies posts related to stress, anxiety, or depression, providing insight into online mental health discourse.

Flight Delay Prediction

[\[Code\]](#)

Developed a machine learning model using XGBoost to forecast flight delays based on historical flight and weather data. Achieved improved prediction accuracy by performing extensive feature engineering and hyperparameter tuning.

Basketball Shot Predictor

[Code]

Created a statistical model using linear regression to predict shot success based on player movement, court position, and historical performance data. Demonstrated data pre-processing, visualization, and sports analytics techniques.

International Hackathons

Azure AI Developer Hackathon [Hackathon Link]

Project: **Legal Hearing Summarizer** – [Project Link]

Global AI Agents League [Hackathon Link]

Project: **BlogPilot AI** – [Project Link]

Awards & Honors

Ignite Final Year Project Funding

2024

Final Year Project on **DevOps and Scalable Kubernetes Deployment** was selected and funded by Ignite.

Top Rated Freelancer on Upwork

2022

Recognized as a Top Rated freelancer for consistent client satisfaction, high-quality project delivery, and maintaining a strong job success score.

University Merit Scholarship

2020–2023

Received partial merit scholarship for five consecutive semesters by securing **first position** in the entire class.

Government of Sindh Award

2017

Awarded a monetary prize for achieving an **A+ Grade** in Matriculation.

Certifications

AWS Certified Cloud Practitioner

Apr 2023

[View Certificate](#)

Hands-on Introduction to Linux Commands and Shell Scripting

Mar 2023

[View Certificate](#)

NodeJS - The Complete Guide (MVC, REST APIs, GraphQL, Deno)

Nov 2022

[View Certificate](#)

py4e101x: Programming for Everybody (Getting Started with Python)

Jul 2020

[View Certificate](#)

Technologies

- **Machine Learning & AI:** scikit-learn, TensorFlow, Hugging Face, LangChain, Pinecone, CrewAI, OpenCV
- **Cloud & DevOps:** AWS Lambda, S3, EC2, DynamoDB, API Gateway, CDK, CodePipeline, Docker, GitHub Actions
- **Backend Development:** Node.js, TypeScript, Express.js, Nest.js, Django, Flask, FastAPI